Alari Ilves

Senior AI/ML Software Engineer

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SUMMARY

Senior AI/ML Software Engineer with 10+ years of experience designing and delivering production-grade intelligent systems across TravelTech, EdTech, and AI Developer Platforms. Expert in building end-to-end AI solutions including LLM applications, NLP systems, Retrieval-Augmented Generation (RAG), machine learning pipelines, scalable microservices, and cloud-native ML infrastructure, transforming large datasets into intelligent automation and decision systems. Proven track record developing high-impact AI products that process millions of events, enable real-time insights, and power enterprise platforms with reliable, scalable, and explainable machine learning capabilities.

WORK EXPERIENCE

Senior AI/ML Software Engineer

Yalantis | Warszawa, Poland | Nov 2021 – Mar 2026

● **Travel & Hospitality AI Automation Platform** — intelligent voice concierge system:
Designed a large-scale conversational AI platform that automates guest requests, reservations, concierge workflows, and hotel service orchestration across global hospitality networks.

● **AI Conversational Intelligence Layer** — multilingual voice interaction and service request understanding:
Built conversational AI pipelines using **Python**, **PyTorch**, **Transformers**, **OpenAI API**, **HuggingFace**, **spaCy**, and speech processing models to enable natural voice-driven interactions between guests and hotel services.

● **Machine Learning Personalization Engine** — predictive service routing and recommendation models:
Developed ML pipelines using **TensorFlow**, **Scikit-learn**, **XGBoost**, **NumPy**, and **Pandas** to predict guest preferences, prioritize requests, and generate personalized concierge responses.

● **Semantic Data Retrieval & Knowledge Intelligence Layer** — contextual service intelligence and guest profile search:
Implemented intelligent search pipelines using **RAG**, **LangChain**, **Pinecone**, **FAISS**, and **Elasticsearch** to retrieve hotel services, policies, and guest history during real-time AI interactions.

● **Scalable Backend Microservices Architecture** — distributed AI orchestration and asynchronous service pipelines:
Built high-performance backend services using **FastAPI**, **gRPC**, **Celery**, **Redis**, **Kafka**, and **Node.js**, enabling thousands of concurrent concierge interactions.

● **Dataset Engineering & Data Pipelines** — hospitality interaction data processing and feature engineering:
Designed ETL and dataset preparation pipelines using **Apache Airflow**, **Spark**, **SQL**, and **PostgreSQL** to process booking logs, guest requests, and service interactions into structured ML training datasets.

● **Cloud Hosting & AI Infrastructure** — global AI service deployment and scalable inference environment:
Deployed AI infrastructure on **AWS** using **EC2**, **S3**, **Lambda**, **SageMaker**, **RDS**, **Docker**, **Kubernetes**, **Terraform**, and **GitHub Actions**, enabling elastic scaling and multi-region reliability.

● **Real-Time AI Integration Layer** — communication between AI services and hotel systems:
Implemented event-driven integrations with PMS, CRM, and reservation systems using **REST APIs**, **GraphQL**, **WebSockets**, and message streaming pipelines.

● **Operational Analytics & Visualization** — hospitality intelligence dashboards:
Built monitoring and analytics dashboards using **Power BI**, **React**, **D3.js**, **Prometheus**, and **Grafana** to visualize service performance, response latency, and guest satisfaction metrics.

Experienced AI/ML Software Engineer

Eremenko & Polomani | Tallinn, Estonia | May 2018 – Oct 2021

● **Travel Industry ML Platform** — automated booking email intelligence system:
Built a machine learning platform that analyzes travel booking emails, extracts structured travel information, and generates automated hotel allocation suggestions.

● **NLP Email Processing Pipeline** — multilingual communication understanding:
Developed language detection, translation, and intent classification pipelines using **NLTK**, **spaCy**, **Transformers**, and **Sentence-Transformers** to process client booking communications.

● **Machine Learning Decision Models** — automated hotel allocation prediction and event detection:
Implemented ML pipelines using **LightGBM**, **XGBoost**, **TensorFlow**, and **Scikit-learn** to detect scheduling conflicts and optimize hotel allocations.

● **Backend Automation Services** — booking workflow orchestration and asynchronous processing:
Designed microservices using **FastAPI**, **RabbitMQ**, **Celery**, and **Redis** to automate booking email processing pipelines.

● **Dataset Processing Pipelines** — structured dataset generation from mailbox exports:
Implemented preprocessing pipelines using **Pandas**, **NumPy**, **BeautifulSoup**, and **Apache Airflow** to transform raw email content into ML training datasets.

● **Cloud Hosting & ML Deployment** — scalable machine learning infrastructure for booking intelligence:
Deployed ML pipelines and inference services on **Microsoft Azure** using **Azure Machine Learning**, **Azure Kubernetes Service**, **Azure Functions**, **Azure Blob Storage**, and **Azure DevOps** to support scalable processing of booking requests.

AI/ML Software Engineer

Terratra | Maardu, Estonia | Sep 2015 - Apr 2018

● **EdTech AI Evaluation Platform** — automated grading and feedback generation system for educational institutions:
Developed an AI-powered grading engine that evaluates assignments, generates explainable feedback, and integrates with learning management systems.

● **RAG-Based Evaluation Intelligence** — rubric grounded scoring and explainable grading:
Implemented evaluation pipelines using **RAG**, **LangGraph**, **Vector Search**, **LLMs**, and **Prompt Engineering** to connect grading decisions with course materials and evaluation policies.

● **Multi-Modal AI Assessment Models** — automated analysis of essays, code, math, and speech:
Developed evaluation models using **PyTorch**, **Transformers**, **Speech Recognition**, and **Code Execution Sandboxes** to support diverse student submission formats.

● **AI Workflow Orchestration** — distributed grading pipelines for high concurrency workloads:
Built event-driven grading workflows using **Kafka**, **Redis**, **PostgreSQL**, and **Microservices Architecture** to process thousands of submissions concurrently.

● **Cloud Hosting & Scalable AI Infrastructure** — production deployment for grading pipelines:
Deployed AI services on **Google Cloud Platform** using **GKE**, **Vertex AI**, **Cloud Run**, **Cloud Storage**, **Docker**, **Kubernetes**, and **Terraform** to ensure scalable grading workloads.

● **LMS Integration Layer** — automated grade publishing and feedback delivery:
Built secure API integrations with LMS platforms using **REST APIs**, **OAuth**, and **Webhooks** for automated grading and feedback distribution.

AI Software Engineer

Golems GABB | Tallinn, Estonia | Jul 2013 – Aug 2015

- **AI Developer Automation Platform** — autonomous AI system that converts engineering tasks into production-ready code: Built an AI-powered engineering assistant capable of analyzing repositories, generating implementation plans, and producing ready-to-merge pull requests.
- **Multi-Agent AI Architecture** — collaborative AI developer agents:
Implemented multi-agent orchestration pipelines using **LLM Multi-Agent Systems**, **AutoGen**, **LangChain**, and tool-based reasoning frameworks to coordinate code generation workflows.
- **Repository Intelligence Engine** — large-scale codebase understanding and dependency reasoning:
Developed repository analysis pipelines using **Code Embeddings**, **Vector Search**, **AST Parsing**, and **Graph-based dependency analysis** to allow AI agents to reason about complex software architectures.
- **AI Code Generation Services** — automated software implementation and testing pipelines:
Built code generation services using **Python**, **LLM APIs**, **Unit Testing Frameworks**, and **Static Analysis Tools** to produce maintainable AI-generated code.
- **DevOps Integration Layer** — automated engineering workflows connected to development tools:
Integrated the AI agent with **GitHub**, **GitLab**, **JIRA**, and **CI/CD pipelines** to enable automated issue analysis, branch creation, and pull request generation.
- **AI Validation & Quality Assurance Systems** — reliability and safety for generated code:
Implemented evaluation pipelines using **Prompt Validation**, **Test Automation**, **Bug Detection Models**, and **Security Scanning** to ensure production-ready code outputs.

SKILLS

AI / Machine Learning: Python, PyTorch, TensorFlow, Scikit-learn, XGBoost, HuggingFace Transformers, NLP, LLMs, Retrieval-Augmented Generation (RAG), LangChain, LangGraph, AutoGen

Data Processing & Engineering: Pandas, NumPy, Apache Airflow, Apache Spark, Feature Engineering, Dataset Pipelines, Data Preprocessing

Search & Knowledge Systems: Vector Databases, Pinecone, FAISS, Elasticsearch, Knowledge Graphs, Semantic Search

Backend & Microservices: FastAPI, Node.js, gRPC, REST APIs, Celery, Kafka, RabbitMQ, Redis

Databases & Storage: PostgreSQL, MongoDB, SQL, Data Lakes

Cloud Hosting, Deployment & MLOps: AWS (EC2, S3, Lambda, SageMaker, RDS, API Gateway), Microsoft Azure (Azure ML, Azure Kubernetes Service, Azure Functions, Azure Blob Storage, Azure DevOps), Google Cloud Platform (GKE, Vertex AI, Cloud Run), Docker, Kubernetes, Terraform, CI/CD (GitHub Actions, Azure DevOps), ML Model Serving, Scalable AI Inference Infrastructure

Monitoring & Observability: Prometheus, Grafana, OpenTelemetry

Visualization & Frontend: React, Power BI, D3.js, Real-Time Dashboards

EDUCATION

Bachelor's Degree in Computer Science
University of Tartu Narva College
Sep 2010 - Jun 2013

PROJECTS

AI Voice Concierge for Hotels and Travel Operators

https://delight.ai

An AI-powered platform was implemented to streamline hotel booking workflows and guest service operations in the travel and hospitality industry. The system combines machine learning, large language models, conversational AI, and predictive analytics to automate guest communication, process booking inquiries, and manage concierge services. Requests from email, voice, phone, or mobile channels are automatically analyzed to detect language, classify intent, and extract key booking details. The platform then checks availability, identifies events or peak periods, and recommends optimal hotel allocations or service actions while generating response suggestions for staff. Integrated with core hospitality systems—including PMS, reservation platforms, POS, CRM, ERP, and smart-room technologies—it creates a unified operational layer that enables real-time coordination across departments. By leveraging guest profiles and historical data for personalization and demand forecasting, the solution reduces response times from minutes to seconds, supports multilingual interactions at scale, and significantly improves operational efficiency and guest experience.

AI-Driven Automated Grading and Feedback Platform for Modern Education

https://www.braintrust.dev

An AI-powered grading system was developed to automate and standardize student assessment in digital learning environments. Using large language models and Retrieval-Augmented Generation (RAG), the platform evaluates assignments against structured rubrics and generates clear, explainable feedback. It supports multiple submission types—including essays, code, math, and audio—while referencing indexed course materials, rubrics, and policies to justify scores and ensure grading consistency. Integrated with Learning Management Systems (LMS), the system automates grading, feedback delivery, and regrade workflows. A human-in-the-loop review process handles low-confidence cases to maintain fairness and accuracy. Built on scalable and secure infrastructure, the platform processes large volumes of submissions efficiently, reducing manual grading workload while improving feedback quality, transparency, and learning outcomes.